

Project Design Phase-II
Solution Requirements (Functional & Non-functional)

Date	1 October 2026
Team ID	6ab6292ddcdae33acb3906da
Project Name	Import Data Using Transform Maps
Maximum Marks	4 Marks

Functional Requirements:

Following are the functional requirements of the proposed solution.

FR No.	Functional Requirement (Epic)	Sub Requirement (Story / Sub-Task)
FR-1	Data Source Definition	• Define a Data Source record (file, JDBC, REST or LDAP) • Upload Excel / CSV / XML files as attachments • Configure file format, sheet number and header row
FR-2	Import Set Table Creation	• Load raw data into a staging (Import Set) table • Auto-create staging table columns from source headers • Preview a sample of the loaded rows
FR-3	Transform Map Creation	• Create a Transform Map linking the Import Set table to the target table • Set the run order and enable/disable the map • Support multiple Transform Maps per Import Set table
FR-4	Field Mapping	• Auto-map fields with matching names • Manually map source fields to target fields • Map reference fields using Choice action and Reference value field • Set coalesce on key fields to update existing records
FR-5	Data Transformation Logic	• Apply source-script field mappings to modify values • Use onStart, onBefore, onAfter and onComplete transform scripts • Use Field Maps to set default or constant values

FR No.	Functional Requirement (Epic)	Sub Requirement (Story / Sub-Task)
FR-6	Data Validation and Cleansing	• Skip or ignore rows with missing mandatory values • Validate formats such as email, date and phone number • Handle duplicate records through coalescing
FR-7	Import Execution	• Run the transform manually from the Import Set • Schedule imports to run automatically (Scheduled Import) • Reuse the same Import Set for repeated loads
FR-8	Import Monitoring and Error Handling	• View inserted, updated, skipped and error counts • Review the Transform History and Import Log • Display error messages for failed rows to enable correction and re-run

Non-functional Requirements:

Following are the non-functional requirements of the proposed solution.

FR No.	Non-Functional Requirement	Description
NFR-1	Usability	Guided, easy-to-follow import steps with an auto-map option and a clear mapping interface, so non-technical admins can configure imports with minimal training.
NFR-2	Security	Role-based access (e.g. import_admin, import_transformer) controls who can create or run imports; sensitive data is protected and import activity is audited.
NFR-3	Reliability	Imports complete consistently without data loss or corruption; failed rows are logged and do not stop the processing of valid rows.
NFR-4	Performance	Large files are processed efficiently in batches, and a standard import of several thousand records completes within an acceptable time window.
NFR-5	Availability	Scheduled imports run reliably at their configured times, and the import feature remains available during normal operating hours with minimal downtime.

FR No.	Non-Functional Requirement	Description
NFR-6	Scalability	The solution handles growing data volumes and additional data sources or Transform Maps without redesign or performance degradation.